



AEO “Nazarbayev Intellectual Schools”

CHEMISTRY

Grade 12

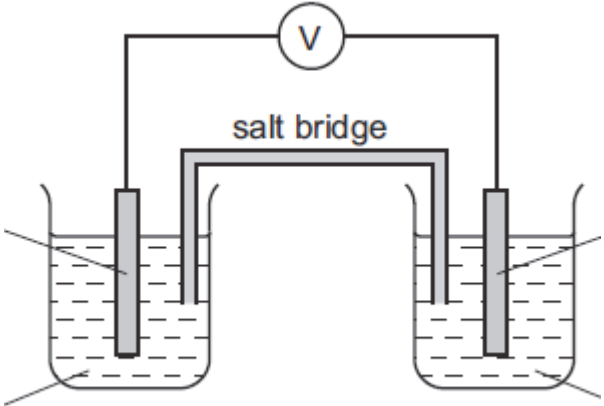
Paper 3

May 2025

MARK SCHEME

Maximum Mark: 30

This document consists of **4** printed pages.

Question	Answer	Mark	Additional Guidance
1(a)	$3\text{Pb} + 2\text{MnO}_4^- + 8\text{H}^+ = 3\text{Pb}^{2+} + 2\text{MnO}_2 + 4\text{H}_2\text{O}$	[1]	Must be balanced
1(b)(i)	 M1 two beakers connected by salt bridge and Voltmeter M2 for correct half-cell $\text{Pb}/\text{Pb}^{2+}(\text{aq})$ M3 for correct half-cell $\text{MnO}_4^-(\text{aq}), \text{H}^+(\text{aq})/\text{MnO}_2$ and Pt	1 1 1 [3]	M3 is not accepted without H^+ ion
1(b)(ii)	+ 1.80V	[1]	Accept 1.8
1(c)	From pink/brown solution to brown ppt	[1]	Accept colourless solution with brown ppt

Question	Answer	Mark	Additional Guidance																								
1(d)(i)	<table><tr><th>concentration of $\text{MnO}_4^- (\text{aq}) / \text{mol dm}^{-3}$</th><th>cell potential / V</th><th>Log $[\text{MnO}_4^- (\text{aq})]$</th></tr><tr><td>$5.00 \times 10^{-1}$</td><td>1.794</td><td>-0.30</td></tr><tr><td>1.00×10^{-1}</td><td>1.780</td><td>-1.00</td></tr><tr><td>4.00×10^{-2}</td><td>1.773</td><td>-1.40</td></tr><tr><td>1.00×10^{-2}</td><td>1.761</td><td>-2.00</td></tr><tr><td>5.00×10^{-3}</td><td>1.755</td><td>-2.30</td></tr><tr><td>2.00×10^{-3}</td><td>1.747</td><td>-2.70</td></tr><tr><td>8.00×10^{-4}</td><td>1.739</td><td>-3.10</td></tr></table>	concentration of $\text{MnO}_4^- (\text{aq}) / \text{mol dm}^{-3}$	cell potential / V	Log $[\text{MnO}_4^- (\text{aq})]$	5.00×10^{-1}	1.794	-0.30	1.00×10^{-1}	1.780	-1.00	4.00×10^{-2}	1.773	-1.40	1.00×10^{-2}	1.761	-2.00	5.00×10^{-3}	1.755	-2.30	2.00×10^{-3}	1.747	-2.70	8.00×10^{-4}	1.739	-3.10		
	concentration of $\text{MnO}_4^- (\text{aq}) / \text{mol dm}^{-3}$	cell potential / V	Log $[\text{MnO}_4^- (\text{aq})]$																								
	5.00×10^{-1}	1.794	-0.30																								
	1.00×10^{-1}	1.780	-1.00																								
	4.00×10^{-2}	1.773	-1.40																								
	1.00×10^{-2}	1.761	-2.00																								
	5.00×10^{-3}	1.755	-2.30																								
	2.00×10^{-3}	1.747	-2.70																								
	8.00×10^{-4}	1.739	-3.10																								
	Correctly calculated values	1																									
All data to 2 decimal places	1																										
	[2]																										
1(d)(ii)	Linear scales used that ensure that when plotted the points occupy at last half of the available grid	1 1 [2]																									
1(d)(iii)	Range 0.150-0.250 OR Any correct calculated value according to students' graph	 [1]																									
1(e)	Higher than Ecell value /1.80 If concentration of Pb^{2+} decrease equilibrium shift to the right.	1 1 [2]																									
1(f)	KCl will form ppt with Pb^{2+}	[1]																									

Question	Answer	Mark	Additional Guidance
1(g)	Pb ⁴⁺ better oxidizing agent/Pb ⁺² weaker reducing agent/ higher electrode potential value	1	ORA
	Ecell negative/ reaction not feasible	1 [2]	OWTTE
2(a)(i)	Propanoic acid/ CH ₃ -CH ₂ -COOH	[1]	
2(a)(ii)	2CH ₃ CH ₂ COOH + MgCO ₃ → (CH ₃ CH ₂ COO) ₂ Mg + CO ₂ + H ₂ O	[1]	
2(a)(iii)	Lime water turns milky	[1]	OWWTE
2(a) (iv)	HCOOCH ₂ CH ₃ / CH ₃ COOCH ₃	[1]	Accept CH ₃ CO-CH ₂ OH/ CH ₃ -CH(OH)-CHO
2(b)(i)	Oxidizing agent: K ₂ Cr ₂ O ₇ / KMnO ₄ Conditions: H ⁺ / H ₂ SO ₄ heat/warming/T	1	Accept names of compounds
		1 1 [3]	Accept acidified
2(b)(ii)	Depends on candidate answer.		
	Orange to green/ pink to colourless	[1]	
2(b)(iii)	Propan-2-ol/ CH ₃ -CH(OH)-CH ₃	[1]	
2(c)	2,4 -DNPH	[1]	
2 (d)	No change with Q	1	
	Silver mirror with M	1 [2]	
2 (e)(i)	NaBH ₄ /LiAlH ₄	[1]	
2 (e) (ii)	CH ₃ -CO-CH ₃ + 2[H] → CH ₃ -CH(OH)-CH ₃	[1]	